

KATHMANDU UNIVERSITY

Course of Study

Course Title: Scientific Computing

Level: Undergraduate

Course Code: DSMA 225

Credit: 3

Group: Data Science (II Year - II Semester)

Prerequisites: Calculus/Linear Algebra/Basic Programming (Python/MATLAB/GeoGebra)

Lecture: 45 Hours

Course Description

This course introduces the fundamental concepts scientific computing and scientific methods used to solve the mathematical problems arising in science, engineering and data science applications. The course focuses on the numerical algorithms for solving nonlinear equations, systems of linear equations, interpolation, data modeling, numerical differentiation and integration, ordinary differential equations and optimization problems. Students will implement these algorithms using computational tools such as Python or MATLAB. The course emphasizes both theoretical foundation and computational skills relevant to data science and scientific applications.

Course Objectives:

- To develop numerical algorithms for solving mathematical problems arising in science, engineering and data science.
- To analyze the numerical errors and convergence properties of the algorithms.
- To implement numerical algorithms using computational tools.
- To develop computational skills and analytical skills for solving real-world problems through min-projects.

Computational and Graphic Tools

Python/MATLAB/GeoGebra

Course Contents

Unit 1: Scientific Computing and Numerical Errors

[5 hours]

- 1.1 Scientific computing in science and engineering.
- 1.2 Sources of computational errors
- 1.3 Approximations and significant figures
- 1.4 Absolute and relative errors
- 1.5 Round-off errors and truncation errors
- 1.6 Error propagation and general error formula

Unit 2: Solution of Nonlinear Equations

[5 hours]

- 2.1 Non-linear equations and solutions

- 2.2 Bisection method and error bound
- 2.3 Regula-Falsi method
- 2.4 Newton–Raphson method
- 2.5 Secant method
- 2.6 Fixed-point iteration method and convergence criteria
- 2.7 Order of convergence of iterative methods

Unit 3: Numerical Linear Algebra

[5 hours]

- 3.1 System of linear equations
- 3.2 LU decomposition
- 3.3 Tri-diagonal system and Thomas Algorithm
- 3.4 Iterative methods: Jacobi method, Gauss–Seidel method
- 3.5 Condition number and ill-conditioned systems
- 3.6 Power method for Eigen-value and Eigen-vector

Unit 4: Interpolation and Curve Fitting

[8 hours]

- 4.1 Polynomial interpolation
 - Finite differences: forward difference, backward difference, central difference
 - Interpolation with equally spaced data set: Newton forward and backward
 - Interpolation with unequally spaced data set: Lagrange interpolation formula
- 4.2 Spline interpolations: Linear, quadratic and cubic
- 4.3 Curve fitting using least square method
 - Straight line fitting and polynomial fittings
 - Exponential and power function fittings

Unit 5: Numerical Differentiation and Integration

[5 hours]

- 5.1 Numerical differentiation based on Newton’s forward and backward interpolation formulas
- 5.2 Newton Cotes integration formula
 - Trapezoidal rule, Simpson’s 1/3 and 3/8 rules
- 5.3 Gaussian quadrature: Gauss–Legendre integration formula

Unit 6: Numerical Solution of Ordinary Differential Equations (ODEs)

[5 hours]

- 6.1 Initial value problems
 - Solution of first order equations: Taylor’s series method, Euler’s method, Runge-Kutta methods (Second and fourth order)
- 6.2 Solution of system of first order ODEs using Euler’s methods and Runge-Kutta second order method

Unit 7: Computational Laboratory

[12 Hours]

- 7.1 Root finding methods: Bisection, Newton-Raphson
- 7.2 Interpolation: Newton’s Forward, Lagrange Interpolation
- 7.3 Curve fitting: Straight line, Exponential and power functions
- 7.4 Solution of linear system: Thomas Algorithm, Gauss-Seidel
- 7.5 Numerical integration: Trapezoidal rule, Simpson’s 1/3 rule, Gauss-Legendre integration
- 7.6 Numerical Solutions of ODEs: Euler’s methods, Runge-Kutta methods

Text Books

1. Numerical Methods for Engineers, Steven C. Chapra , Raymond P. Canale, McGraw-Hill
2. Introductory Methods of Numerical analysis, S. S. Sastry, PHI Learning Private Limited, New Delhi

Reference Books

1. Numerical Methods Using MATLAB, John H. Mathews and Kurtis D. Fink, Pearson Prentice Hall
2. Numerical Methods in Engineering & Science, B. S. Grewal, Khanna Publishers, India
3. Numerical Python: Scientific Computing and Data Science Applications with Python, Robert Johansson, Apress.

Evaluation Modality:

1. In-Semester Evaluation [40 marks]

2. End Semester Evaluation [60 marks]

- 20 marks for Objective (10 Q. $\times$ 1 = 10 marks for fill in the blank questions and 10 Q. $\times$ 1 = 10 marks for multiple choice questions)
- 40 marks for Subjective questions (2 Q. $\times$ 8 = 16 marks for long answer questions and 6 Q. $\times$ 4 = 24 marks for short answer questions.)